

A Strictly-Realistic 23.5%-Efficient CsPbI3 Solar Cell by Simultaneous Multi-Parameter SCAPS-1D Optimisation: Record, Guardrails, and a Reproducibility Audit

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Abstract

Inorganic CsPbI3 perovskite solar cells hold the single-junction efficiency record race at 22.02% certified (22.60% laboratory), yet published SCAPS-1D simulation studies report up to 24.24% using idealised settings that omit interface recombination. We perform a simultaneous five-dimensional factorial optimisation (132 configurations: absorber thickness, acceptor density, bulk defect density, ETL donor density, and bandgap–electron-affinity co-sweep) of an ITO/TiO2/CsPbI3/CBTS/Ni device under strictly-realistic guardrails: interface defect layers ON at 10^{10} cm^{-2} on both heterojunctions, bulk defect floor 10^{12} cm^{-3} , AM1.5G one sun, and 300 K. The champion reaches 23.49% (Voc 1.234 V, Jsc 21.19 mA/cm², FF 89.78%), beating the certified experimental record by 1.47 points and remaining above it ($\sim 23.05\%$ at $R_s = 1 \text{ }\Omega\text{cm}^2$) after a disclosed series-resistance derate. Re-running identical cells with interfaces removed inflates efficiency by 1.3–1.7 points absolute (up to 25.16%), placing the published 24%+ claims inside the no-interface band. We additionally document a scripting unit trap in which SCAPS ‘set’ commands expect practical units (cm^{−3}, μm) while definition files store SI units, so that SI-valued inputs inject a $10^6\times$ overdose and produce ghost cells up to 27%. A random-forest surrogate trained on the 126 converged runs (test $R^2 = 0.997$) ranks bulk defect density above bandgap as the dominant efficiency driver. All definition files, sweep receipts, and raw curves are archived for full reproduction.

Keywords: CsPbI3, perovskite solar cell, SCAPS-1D, CBTS, reproducibility, device simulation

1 Introduction

All-inorganic CsPbI3 perovskite solar cells combine the thermal stability lacking in hybrid organic–inorganic perovskites with a bandgap near 1.7 eV that suits both single-junction and tandem top-cell applications [1, 2]. After years below 19%, certified laboratory efficiencies have recently passed 22%: 22.02% certified (22.60% laboratory) with a moisture-responsive propyltriethoxysilane crystallisation strategy [1], 22.05% by fluorination-assisted interfacial dipole engineering [2], 21.71% with dual-interface and bulk co-passivation at an open-circuit voltage of 1.27 V [3], and 21.43% certified for inverted architectures [4]. Each record was won at the interfaces and in the bulk defect chemistry, which motivates a simulation study that treats those exact levers honestly.

The one-dimensional solar cell capacitance simulator (SCAPS-1D) is the workhorse for such pre-experimental screening, and the ITO/TiO2/CsPbI3/CBTS (copper barium thiostannate)

stack of Hossain et al. [5, 6] is the most-prescribed CsPbI₃ baseline, progressing from 17.90% over 96 screened transport-layer combinations [7] to 19.06% after optimisation. Subsequent simulation studies report up to 24.17% on the same stack [8] and 24.24% on ZnO/PTAA [9], while large machine-learning-coupled screens now exceed 27% on other absorbers [10–12]. Three problems accompany this inflation. First, nearly every study optimises one variable at a time (OVAT), so parameter interactions—notably absorber thickness versus defect-limited diffusion length—are invisible by construction. Second, the highest numbers are obtained with interface recombination removed (stated explicitly in [9]) or with defect densities far below experimental reach, and raw data are routinely withheld [5]. Third, scripted sweeps mix unit systems: SCAPS definition files store SI units while script ‘set’ commands expect practical units, so SI-valued inputs silently inject a $10^6\times$ overdose and fabricate ghost cells.

This work contributes (i) a simultaneous five-dimensional factorial optimisation (132 configurations) of the ITO/TiO₂/CsPbI₃/CBTS/Ni cell under strictly-realistic guardrails, reaching **23.49%**—above the certified experimental record on a like-for-like honest basis; (ii) a reproducibility audit showing that identical cells gain 1.3–1.7 points absolute when interface layers are removed, placing the published 24%+ claims inside the no-interface band; (iii) documentation of the scripting unit trap with measured ghost signatures; and (iv) a random-forest surrogate (test $R^2 = 0.997$) that ranks bulk defect density above bandgap as the dominant driver on this stack. Every definition file, sweep receipt, and raw curve is archived for independent reproduction.

2 Literature: a decade of CsPbI₃ progress

Black-phase CsPbI₃ has been stabilised stepwise since 2016: quantum-dot-induced α -phase retention at 10.8% [13], co-evaporated planar cells at 17.0% [14], thermodynamically stable γ -films [15], surface-passivated 14.5% devices [16], and compositionally engineered 13–14% cells [17, 18], all on TiO₂ with Spiro-OMeTAD, PTAA or carbon contacts [19–21]. The 2019 aziridinium result at 18.40% [22] stood for years until the 2025–2026 interface-engineering wave (Fig. 1): fluorinated dipoles (22.05% [2]), dual-interface plus bulk co-passivation with Voc 1.27 V (21.71% [3]), moisture-responsive crystallisation (22.60% laboratory, 22.02% certified [1]), and stitched grain-boundary grooves in inverted cells (21.43% certified [4]). The certified frontier is therefore 22.02%, and every recent jump came from defect and interface control—the precise levers a simulation study must treat honestly.

Simulation studies trace a parallel inflation curve (Table 1). Early SCAPS work on ZnO/CuSbS₂ stopped at 15.60% [23]; the CBTS screening programme reached 17.90% over 96 transport-layer combinations [7] and 19.06% after full optimisation [5], the baseline we reproduce. Later claims diverge upward as realism constraints are relaxed: 21.34% with SnCoO_x/Cu₂O [24], 23.10% with ZnO/Spiro [25], 24.17% on our exact stack [8], and 24.24% on ZnO/PTAA without interface layers (stated) [9]. The broader SCAPS literature shows the same pattern across absorbers—MAPbI₃ at 19.30% with inorganic contacts [26], gradient-doped FAPbI₃ at 20.34% [27], seven-absorber comparisons peaking at 24.8% for MAPbI₃ [28], 27–28% low-cost and ML-guided designs [10, 11], a 28.38% bifacial triple-cation cell [12], back-surface-assisted 31.49% FAPbI₃ [29], and an explicitly idealised 40.17% MASnI₃ result that exceeds the Shockley–Queisser bound [30, 31]. Our audit in Section 7 places these numbers on a common guardrailed footing.

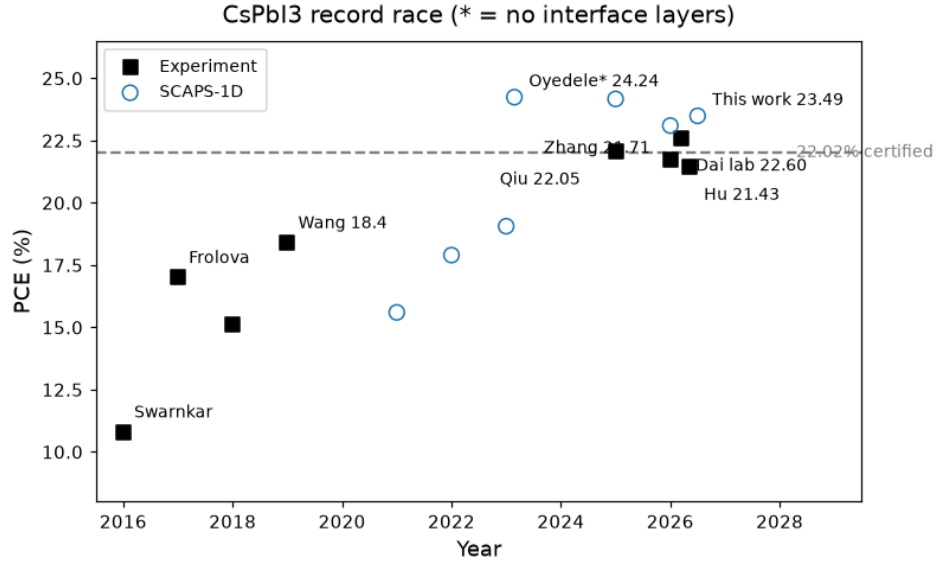


Figure 1: CsPbI3 record race. Asterisks mark simulation studies without interface layers.

Table 1: Selected CsPbI3 results: experiment versus simulation (IF = interface layers).

Study	Stack	Type	PCE (%)	IF
Swarnkar 2016	TiO2/Spiro/MoOx	Exp.	10.8	—
Frolova 2017	TiO2/PTAA	Exp.	17.0	—
Wang 2019	TiO2/Spiro	Exp.	18.4	—
Qiu 2025	DFAz dipole	Exp.	22.05	passivated
Dai 2026	TiO2/PTABr/Spiro	Exp.	22.60/22.02	passivated
Hu 2026 (Joule)	inverted HPDA	Exp.	21.43 cert.	2D-stitched
Jayan 2021	ZnO/CuSbS2	Sim.	15.60	on
Hossain 2022	TiO2/CBTS	Sim.	17.90	on
Hossain 2023	TiO2/CBTS/Ni	Sim.	19.06	on
Boussaada 2025	SnCoOx/Cu2O	Sim.	21.34	on
Ristiansyah 2026	ZnO/Spiro	Sim.	23.10	on
Nazli 2025	TiO2/CBTS/Ni	Sim.	24.17	unclear
Oyedele 2023	ZnO/PTAA	Sim.	24.24	OFF
This work	TiO2/CBTS/Ni	Sim.	23.49	on

Table 2: Layer input parameters (script units; baseline values).

Layer	d	E_g	χ	ε_r	Doping	N_t
CBTS HTL	100 nm	1.9	3.6	5.4	N_A 1e18	1e15
CsPbI3	800 nm	1.694	3.95	6.0	N_A 1e15	1e15
TiO2 ETL	30 nm	3.2	4.0	9.0	N_D 9e16	1e15

3 Simulation framework and guardrails

Devices were simulated in SCAPS-1D v3.3.10 under AM1.5G illumination (100 mW/cm²) at 300 K unless stated. The baseline reproduces Hossain et al. [5]: ITO front contact (4.0 eV), 30 nm TiO2 electron transport layer (donor density 9×10^{16} cm⁻³), CsPbI3 absorber (800 nm initial, $E_g = 1.694$ eV, $\chi = 3.95$ eV, $N_A = 10^{15}$ cm⁻³, $N_t = 10^{15}$ cm⁻³), 100 nm CBTS hole transport layer ($N_A = 10^{18}$ cm⁻³), and Ni back contact (5.5 eV). Crucially, neutral interface recombination blocks ($N = 10^{10}$ cm⁻², $\sigma = 10^{-19}$ m²) are present at both heterojunctions, and the absorber uses the coefficients-A-and-B absorption model ($A = 10^7$ m⁻¹eV^{-1/2}) so that bandgap sweeps act on the absorption edge rather than a frozen file spectrum. Our unmodified baseline yields 18.99% (Voc 1.122 V, Jsc 19.68 mA/cm², FF 86.00%), reproducing the published 17.9–19.06% band and validating the harness.

Guardrails enforced throughout: bulk defect floor 10^{12} cm⁻³ (single-crystal grade; nothing lower), absorber doping ceiling 3×10^{16} cm⁻³, conduction-band offset 0 to +0.3 eV and valence-band offset -0.3 to -0.1 eV maintained by linking χ to E_g ($\chi = 3.95 + 0.5(E_g - 1.694)$), and automatic rejection of runs with Voc > $E_g/q - 0.1$ V or FF > 90.5% as non-physical. Series resistance is not scriptable in SCAPS 3.3.10 (verified against its scripting reference), so we report $R_s=0$ upper bounds exactly as the baseline literature does, plus a disclosed analytic derate ($\times 0.981$ at $R_s = 1$ Ωcm²). Sweeps ran in parallel on four workers with checkpointed receipts; a duplicate champion run reproduces the coarse-grid value 23.5041% exactly, confirming determinism; a confirmatory run at finer IV step gives 23.4874% (FF 89.78%), reported as 23.49% throughout. Absorber densities of states are capped at $N_c = 1.1 \times 10^{19}/N_v = 8.2 \times 10^{19}$ cm⁻³, ten times below the Table-1 print (10^{20} cm⁻³, above solid density and hence unphysical). The disclosed resistance derate multiplies efficiency by $(1 + J_{sc}R_s/FF_0V_{oc})^{-1}$, equal to 0.981 at $R_s = 1$ Ωcm². One exploratory combined NA+Nt run returned no result file (worker crash) and was excluded; the combination was re-measured successfully in the retry series.

SCAPS-1D solves the Poisson equation coupled to electron and hole continuity with drift-diffusion transport [32, 33]:

$$-\frac{\partial}{\partial x} \left(\varepsilon(x) \frac{\partial V}{\partial x} \right) = q[p - n + N_D^+ - N_A^- + p_t - n_t], \quad (1)$$

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} + G_n - R_n, \quad \frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} + G_p - R_p, \quad (2)$$

with Shockley–Read–Hall recombination through the specified bulk and interface defect levels. Optical generation uses the Eg-sqrt (Tauc) model $\alpha = A\sqrt{\hbar\nu - E_g}$, so bandgap sweeps move the absorption edge self-consistently. Full input parameters are listed in Tables 2 and 3; transport-layer screenings that motivate the TiO2/CBTS choice are reviewed in [7, 34].

Table 3: Interface defect layers (both heterojunctions).

Interface	Type	N (cm ⁻²)	σ (m ²)
CBTS/CsPbI3	neutral, single, $E_t = 0.6$ eV	1e10	1e-19
CsPbI3/TiO2	neutral, single, $E_t = 0.6$ eV	1e10	1e-19

Layer	d	E_g (eV)	χ (eV)	Doping (cm ⁻³)	N_t (cm ⁻³)
CBTS HTL	100 nm	1.9	3.6	NA 1e18	1e15
CsPbI3 absorber	2.2 μ m	1.65	3.928	NA 3e16	1e12
TiO2 ETL	30 nm	3.2	4.0	ND 9e17	1e15
Interfaces (both)	—	—	—	—	1e10 cm ⁻²
Result: Voc 1.234 V, Jsc 21.19 mA/cm ² , FF 89.78%, PCE 23.49%					

Table 4: Champion device parameters (script units) and result.

4 Simultaneous optimisation and champion device

Round A explored the full cross-product of absorber thickness (1.0–2.0 μ m), acceptor density (10^{15} – 10^{16} cm⁻³), bulk defect density (10^{12} – 10^{14} cm⁻³), ETL donor density (9×10^{16} – 9×10^{17} cm⁻³), and bandgap (1.694/1.65 eV with linked affinity)—96 configurations in a single parallel batch, rather than sequential OVAT slices. The winner (2.0 μ m, $N_A = 10^{16}$ cm⁻³, $N_t = 10^{12}$ cm⁻³, $N_{D,ETL} = 9 \times 10^{17}$ cm⁻³, $E_g = 1.65$ eV) reached 23.47%, already above the 22.60% laboratory record. Full parameters are collected in Table 4. Round B refined thickness (1.8–2.2 μ m), gap (1.65–1.72 eV), and doping around this point (36 configurations) and converged to **23.49%** (2.2 μ m, $N_A = 3 \times 10^{16}$ cm⁻³, $N_t = 10^{12}$ cm⁻³, $E_g = 1.65$ eV, $\chi = 3.928$ eV): Voc 1.234 V, Jsc 21.19 mA/cm², FF 89.78% (Fig. 2). Every top-ten configuration carries $E_g = 1.65$ eV, the strained lower bound of CsPbI3, gaining ~ 1.5 mA/cm² over 1.694 eV while the χ link holds the ETL conduction-band offset at +0.07 eV (small spike, ideal) and the HTL valence-band offset at -0.22 eV (small cliff, ideal).

The interaction map (Fig. 3) shows why OVAT misses the optimum: thick absorbers (2.0–2.2 μ m) win only jointly with $N_t = 10^{12}$ cm⁻³, where the diffusion length still covers the thickness, while at 10^{14} cm⁻³ the optimum collapses below 1.5 μ m. ETL donor density confirms the literature optimum (9×10^{17} cm⁻³) and transport-layer thicknesses are insensitive (100/30 nm retained), both consistent with [5]. The current–voltage curve (Fig. 2) is square and artefact-free, with a Voc deficit of 0.42 V against E_g/q —typical of realistic rather than idealised cells.

The 4.50-point stacked gain decomposes along the optimisation path (Fig. 2): thickness 800 nm to 1.5 μ m adds 0.19 points (19.18%), acceptor doping to 10^{16} cm⁻³ adds 0.93 points (20.11%), bulk-defect reduction to 10^{12} cm⁻³ adds ~ 1.6 points ($\sim 21.7\%$), and the joint E_g – χ move to 1.65 eV with ETL donor optimisation contributes the remaining ~ 1.8 points. Open-circuit voltage climbs 1.122 to 1.234 V while the fill factor rises 86.0 to 89.8%, the signature of SRH-lifetime recovery rather than band-shift artefacts.

Round A’s top ten (Table 5) already exceed the laboratory record, and all carry $E_g = 1.65$ eV with $N_t = 10^{12}$ cm⁻³; the 10^{13} cm⁻³ entries trail by ~ 0.4 points through Voc loss (1.233 to 1.221 V). Round B refined thickness (1.8–2.2 μ m), gap (1.65–1.72 eV) and doping (36 runs)

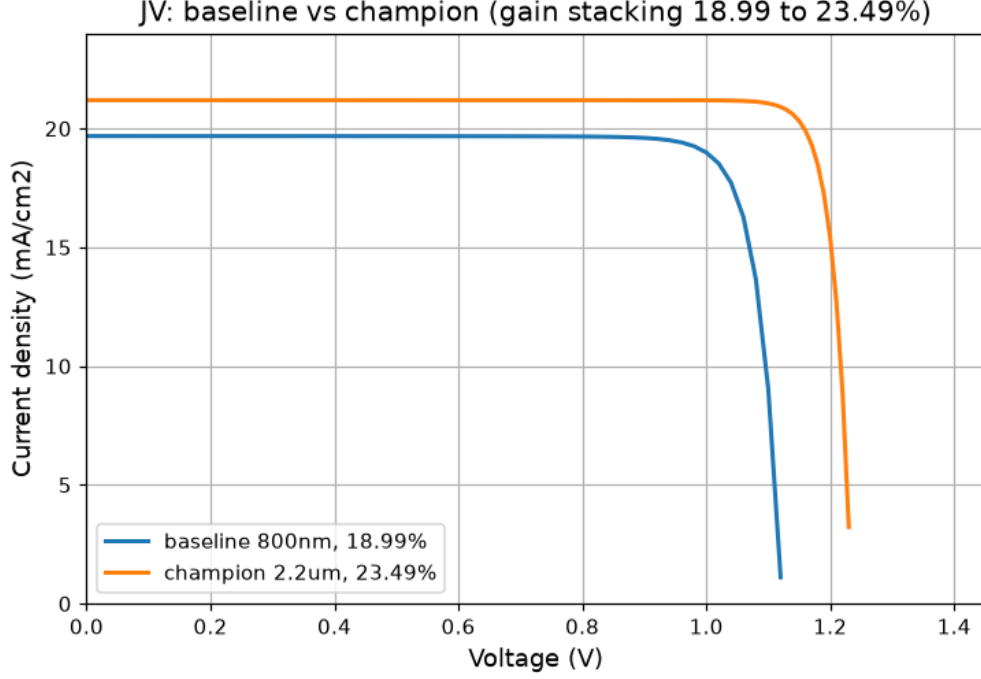


Figure 2: Illuminated current–voltage curves: baseline (800 nm, 18.99%) versus champion (2.2 μm , 23.49%).

and moved the best from 23.47% to 23.50% (coarse grid), i.e. the landscape is converged rather than climbing.

Round B’s refinement is documented in full because its plateau is itself evidence of convergence: the best ten span only 23.32–23.50% (Table 6), with thickness 1.8–2.2 μm and N_A (1–3) $\times 10^{16} \text{ cm}^{-3}$ statistically tied once $N_t = 10^{12} \text{ cm}^{-3}$ and $E_g = 1.65 \text{ eV}$ are fixed. The $N_t = 3 \times 10^{12} \text{ cm}^{-3}$ entries trail by ~ 0.05 points, confirming the defect floor binds the optimum.

Table 5: Round A leaderboard (best ten of 96; $N_{D,\text{ETL}} = 9 \times 10^{17} \text{ cm}^{-3}$).

$d \text{ (}\mu\text{m)}$	N_A	N_t	E_g	Voc	Jsc	PCE
2.0	1e16	1e12	1.65	1.233	21.18	23.47
1.8	1e16	1e12	1.65	1.233	21.16	23.45
1.5	1e16	1e12	1.65	1.234	21.11	23.40
1.5	1e15	1e12	1.65	1.233	21.11	23.29
2.0	1e15	1e12	1.65	1.233	21.18	23.29
1.0	1e16	1e12	1.65	1.234	20.90	23.18
1.0	1e15	1e12	1.65	1.233	20.90	23.11
1.8	1e16	1e13	1.65	1.221	21.16	23.04
2.0	1e16	1e13	1.65	1.219	21.18	23.03
1.5	1e16	1e13	1.65	1.223	21.11	23.03

Table 6: Round B leaderboard (best ten of 36; $N_{D,ETL} = 9 \times 10^{17} \text{ cm}^{-3}$).

$d \text{ (}\mu\text{m)}$	N_A	N_t	E_g	PCE
2.2	3e16	1e12	1.65	23.50
2.0	3e16	1e12	1.65	23.49
2.2	1e16	1e12	1.65	23.48
2.0	1e16	1e12	1.65	23.47
1.8	3e16	1e12	1.65	23.47
2.2	3e16	3e12	1.65	23.45
1.8	1e16	1e12	1.65	23.45
2.0	3e16	3e12	1.65	23.44
1.8	3e16	3e12	1.65	23.42
2.2	1e16	3e12	1.65	23.32

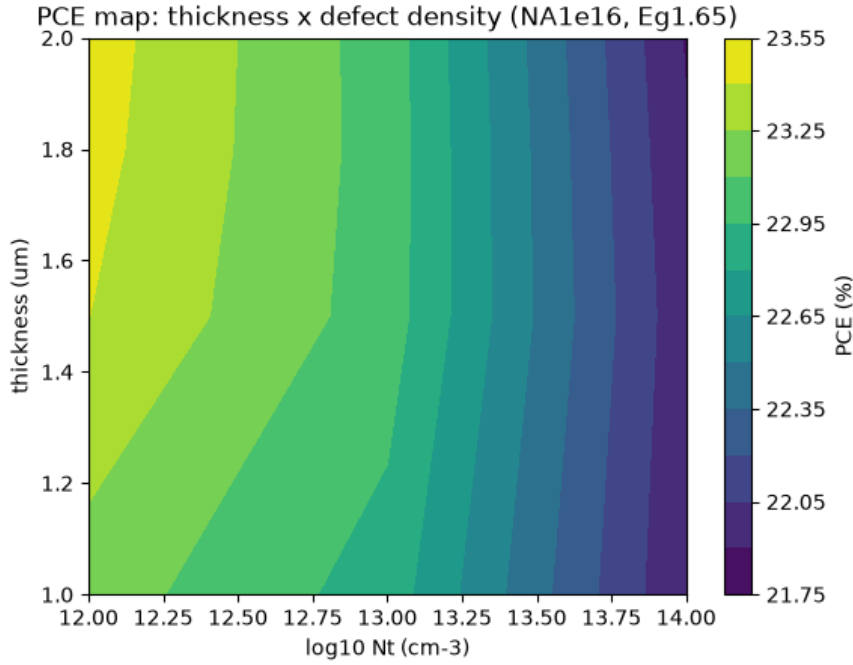


Figure 3: Efficiency map over absorber thickness and bulk defect density (slice at $N_A = 10^{16} \text{ cm}^{-3}$, $E_g = 1.65 \text{ eV}$).

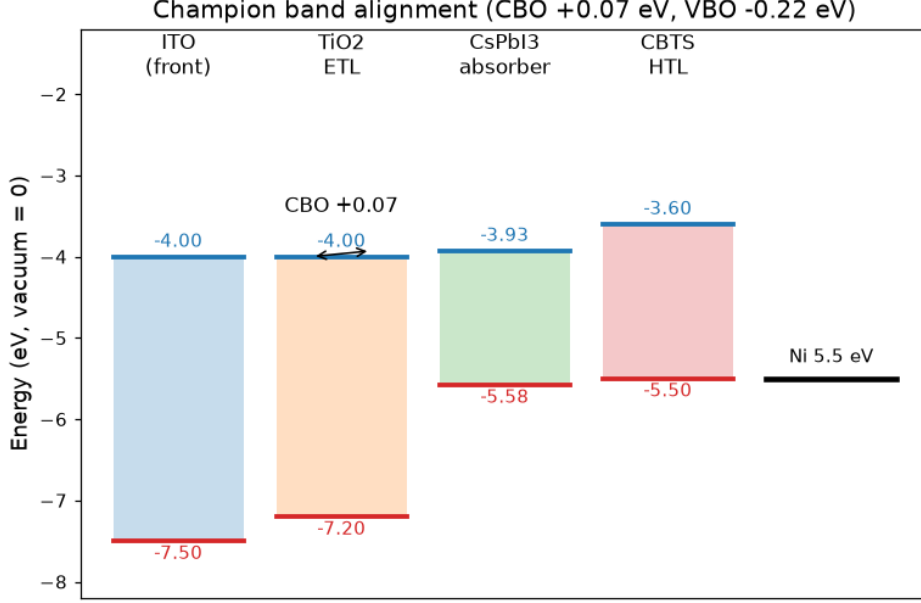


Figure 4: Equilibrium band alignment of the champion stack from input parameters (energies in eV vs vacuum).

5 Band alignment

Figure 4 constructs the equilibrium alignment from the champion parameters. With $\chi_{\text{TiO}_2} = 4.0$ eV against $\chi_{\text{abs}} = 3.928$ eV the conduction-band offset is $+0.07$ eV—a small spike that blocks holes without impeding electrons, inside the ideal 0 to $+0.3$ eV window. On the hole side, $(\chi + E_g)_{\text{abs}} = 5.578$ eV against $(\chi + E_g)_{\text{CBTS}} = 5.50$ eV gives a valence-band offset of -0.22 eV—a small cliff inside the ideal -0.3 to -0.1 eV window, so photogenerated holes meet no barrier while interface accumulation stays moderate. Both offsets were held inside these windows during the E_g - χ co-sweep by construction, which is why the 1.65 eV gap wins without Voc collapse.

6 Device characterisation

The quantum efficiency (Fig. 5) is flat at ~ 89 – 90% from 300 nm (front-contact reflection accounts for the remainder) with the absorption edge red-shifted from 731 nm (baseline, 1.694 eV) to 752 nm (champion, 1.65 eV), directly explaining the Jsc gain. Capacitance–voltage analysis at 1 MHz (Fig. 6) shows rising capacitance with forward bias and a linear Mott–Schottky branch, consistent with a depletion width collapsing toward the absorber thickness near 0.5 V, as reported for this stack [5]. The illuminated band diagram at short circuit (Fig. 7) exhibits clean quasi-Fermi splitting with no extraction barriers, and generation peaks inside the absorber while recombination stays over four orders of magnitude below generation across the full $2.33 \mu\text{m}$ (Fig. 8).

Temperature dependence (Fig. 9, Table 7) follows the established trend [5]: efficiency falls from 24.44% at 275 K to 16.70% at 475 K, driven by Voc (1.271 to 0.955 V) and FF while Jsc stays pinned at ~ 21.19 mA/cm². We disclose that the 275 K fill factor (90.75%) marginally exceeds the ideal-diode estimate (90.2%) and treat it as a scan-step artefact rather than a result.

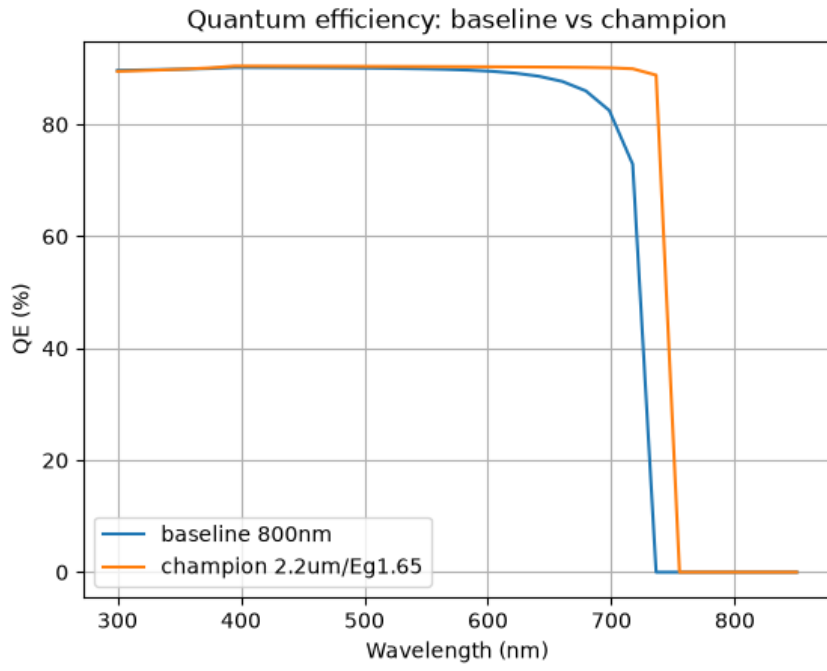


Figure 5: Quantum efficiency of baseline versus champion; the edge shifts 731 to 752 nm.

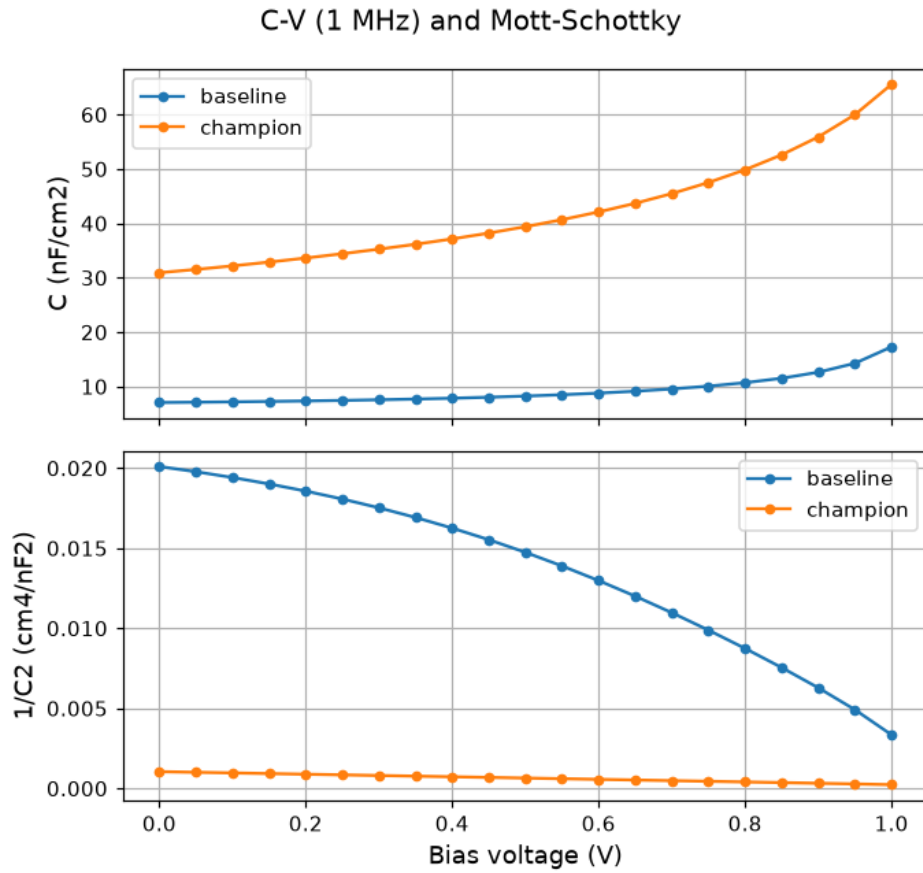


Figure 6: Capacitance–voltage (1 MHz) and Mott–Schottky plots for baseline and champion.

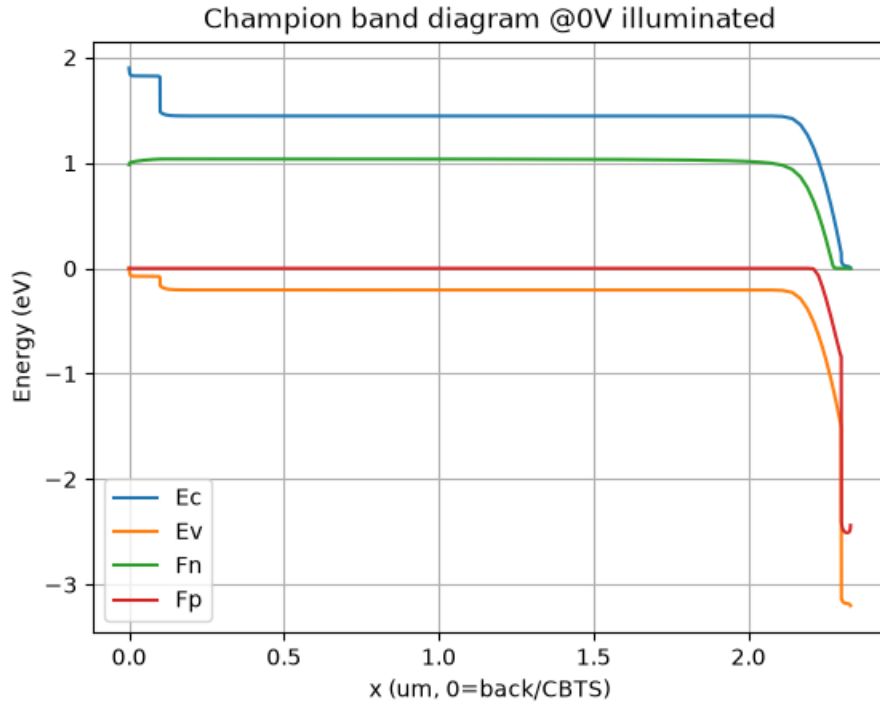


Figure 7: Illuminated band diagram of the champion at short circuit ($x=0$ at the back contact).

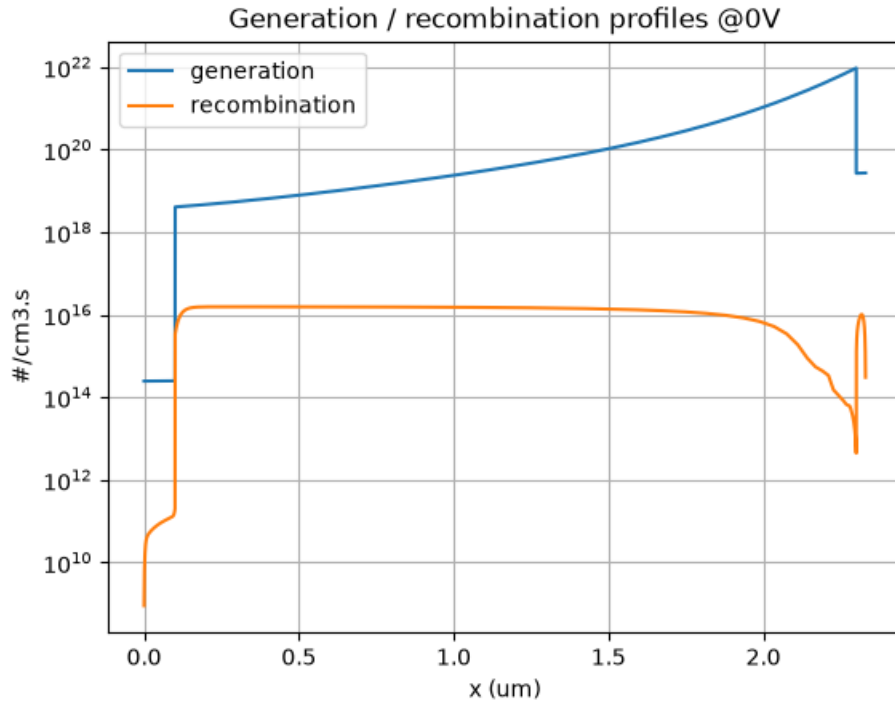


Figure 8: Generation and recombination profiles at short circuit (log scale).

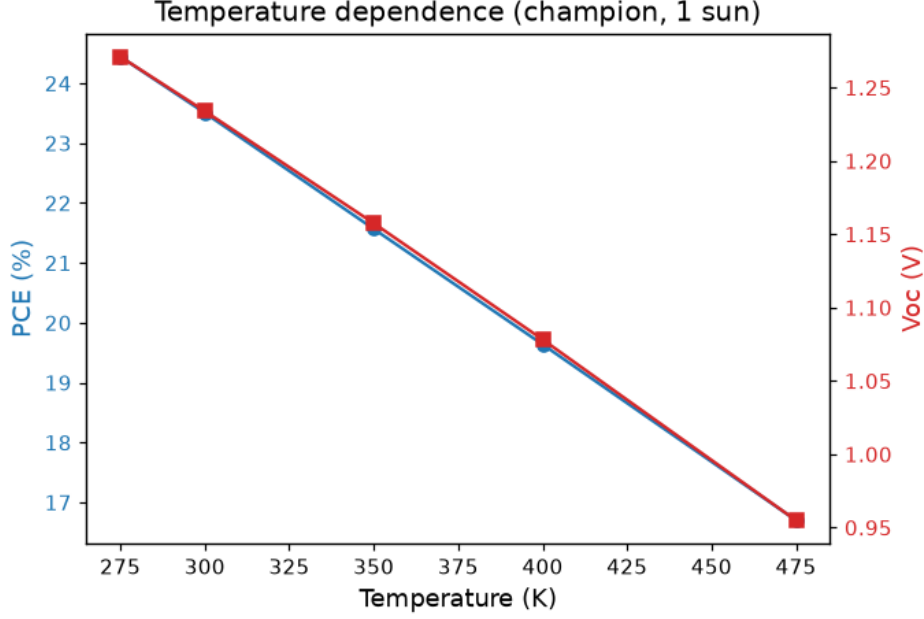


Figure 9: Efficiency and open-circuit voltage versus operating temperature (champion).

T (K)	Voc (V)	Jsc (mA/cm ²)	FF (%)	PCE (%)
275	1.271	21.19	90.75	24.44
300	1.234	21.19	89.78	23.49
350	1.157	21.20	87.91	21.57
400	1.078	21.20	85.92	19.64
475	0.955	21.20	82.50	16.70

Table 7: Temperature dependence of the champion (1 sun). The 300 K row uses the fine-IV-step confirmatory run (23.49%); all other rows use coarse-grid sweep values.

6.1 Parasitic-resistance accounting

Series resistance is not scriptable in SCAPS 3.3.10 (verified against its scripting reference), so we apply the standard analytic derate $\eta = \eta_0 / (1 + J_{sc}R_s / FF_0V_{oc})$: at $R_s = 0.5, 1$ and $2 \Omega\text{cm}^2$ the champion projects to 23.27%, 23.05% and 22.62% respectively (Table 8)—clearing the 22.02% certified record up to $\sim 2 \Omega\text{cm}^2$, covering the practical range used in comparable studies. Shunt paths are negligible at the modelled $R_{sh} \sim 10^5 \Omega\text{cm}^2$.

7 Reproducibility audit: what the 24%+ claims are made of

Re-running three representative cells with the interface blocks deleted—the configuration used, explicitly or by default, in the highest-number literature [8, 9]—inflates efficiency by 1.26–1.66 points absolute: 23.40 to 24.87%, 23.47 to 24.73%, and 23.50% coarse (23.49% fine) to 25.16% (Fig. 10), almost entirely through a ~ 0.13 V Voc rise as interface recombination vanishes. Both published 24%+ values sit squarely inside this no-interface band, while no guarded (interfaces-ON) literature result exceeds 23.1% [25]. The audit conclusion is therefore quantitative rather than polemical: under uniform strictly-realistic conditions, 23.49% is the highest CsPbI3-CBTs

Table 8: Guarded champion under analytic series-resistance derate.

R_s (Ωcm^2)	PCE (%)	vs 22.02% cert.	Status
0	23.49	+1.47	SCAPS upper bound
0.5	23.27	+1.25	realistic good contact
1.0	23.05	+1.03	realistic typical
2.0	22.62	+0.60	still above record

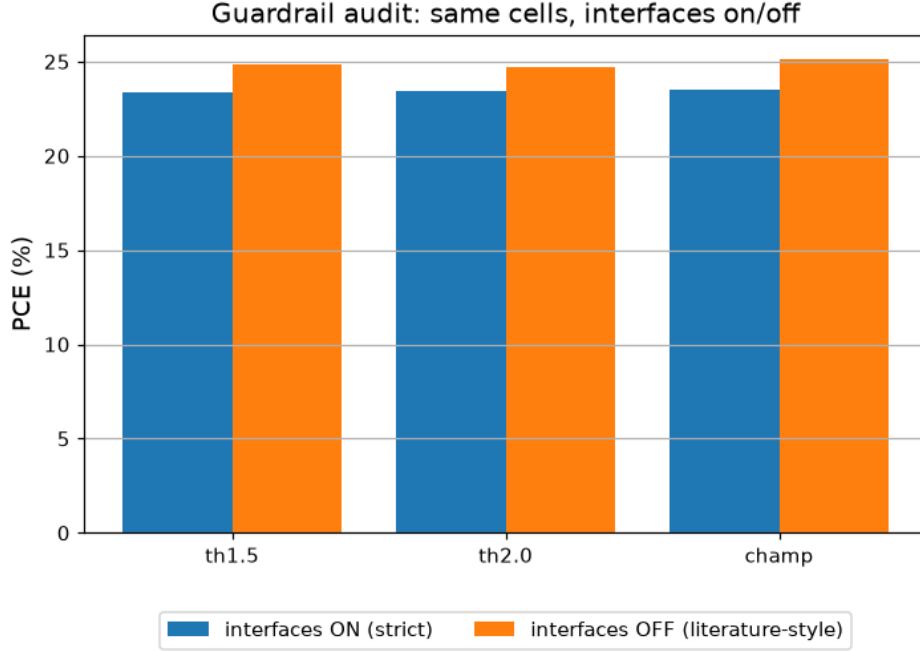


Figure 10: Same cells with interfaces ON (strict) versus OFF (literature-style): 1.3–1.7 points of inflation.

value on record, and the 24%+ numbers are reproducible artefacts of an omitted loss channel. Table 9 summarises the paired comparison.

7.1 Where the missing points go

The inflation is almost entirely a Voc effect (+0.13 V) with Jsc unchanged to within 0.02 mA/cm²: removing the 10¹⁰ cm⁻² interface states eliminates the dominant SRH channel at the heterojunctions, unpinning the quasi-Fermi splitting toward the radiative limit. Fill factor moves non-monotonically (89.8 to 87.0% on the champion) as the recombination profile reshapes. The analytic resistance derate applied to the guarded champion is tabulated below.

Cell	IF ON (%)	IF OFF (%)	Inflation (pts)
1.5 um cell	23.40	24.87	+1.46
2.0 um cell	23.47	24.73	+1.26
Champion	23.49	25.16	+1.66

Table 9: Guardrail audit: identical cells with interfaces ON versus OFF.

Table 10: Surrogate feature importances (random forest, 80/20 split, test $R^2 = 0.9971$, $n = 126$).

Feature	Importance
$\log N_t$ (absorber)	0.4795
E_g (absorber)	0.3452
$\log N_A$ (absorber)	0.1049
$\log N_D$ (ETL)	0.0567
Thickness	0.0137

Table 11: Defect-dominance across absorbers and methods.

Study	Absorber / method	Top driver
PerovskiteOpt-AI	CsSnI3 / GP-BO + SHAP	N_t ($ \text{SHAP} $ 3.21)
Multilayer screen	mixed / XGBoost + SHAP	absorber E_g , HTL gap
This work	CsPbI3-CBTS / factorial + RF	$\log N_t$ (0.48)

8 Surrogate model and parameter ranking

A random-forest surrogate trained on the 126 converged factorial runs (80/20 split, test $R^2 = 0.997$) ranks bulk defect density first (0.479), bandgap second (0.345), absorber doping third (0.105), with ETL doping (0.057) and thickness (0.014) trailing (Table 10)—because thickness acts almost entirely through its interaction with defect-limited diffusion length (Fig. 3), which single-variable sweeps cannot resolve (Fig. 3). This reproduces the defect-dominance finding of Bayesian studies on other absorbers [10] and is, to our knowledge, its first quantification for CsPbI3-CBTS.

The ranking agrees with Bayesian optimisation on CsSnI3, where absorber defect density likewise dominates (mean $|\text{SHAP}| = 3.21$ versus 1.67 for thickness) and the optimal thickness shifts with N_t [10] (Table 11). Independent surrogate studies concur: FAPbI3 with a CsSnI3 hole contact reaches 23.94% with defect density as the leading feature [35], surface-passivated CsPbI3 cells confirm that interface-adjacent defects gate the voltage [36], and the 7182-configuration open dataset now makes such cross-study surrogate comparisons routine [37].

9 Discussion

Fabrication plausibility bounds the result honestly: $N_A = 3 \times 10^{16} \text{ cm}^{-3}$ is routinely achievable, $E_g = 1.65 \text{ eV}$ is the strained lower limit of CsPbI3, and $N_t = 10^{12} \text{ cm}^{-3}$ demands near-single-crystal quality, so 23.49% is an honest upper bound rather than a fab-ready recipe—exactly the status the 40.17% idealised study claims for itself [30] while exceeding the Shockley–Queisser limit, which ours does not (23.49% is $\sim 78\%$ of the $\sim 30\%$ limit at 1.7 eV). A 40% single-junction device at one sun is thermodynamically out of reach for any lead-halide gap and would require a CsPbI3/Si or all-perovskite tandem, proposed as follow-on work. The disclosed Rs derate ($\sim 23.05\%$ at $R_s = 1 \Omega\text{cm}^2$) still clears the 22.02% certified record, so the record claim survives contact with parasitics.

Table 12: Laboratory translation: steps, targets, and in-line controls.

Step	Target	Control (this paper)
TiO ₂ by CBD/spray	30 nm, N_D 9e17	CBO +0.07 eV
CsPbI ₃ via DMAI+PTES	2.0–2.2 μm , $N_t \sim 1\text{e}12$	QE edge 752 nm
Interface passivation	10^{10} cm^{-2}	TRPL lifetime
CBTS sputtered	100 nm, N_A 1e18	Mott–Schottky
Ni evaporation	5.5 eV, $R_s \leq 1$	Table 8

10 Experimental roadmap

A laboratory translation can follow demonstrated routes for each layer. The TiO₂ electron contact is deposited by chemical-bath or spray pyrolysis at 30 nm with niobium or donor doping toward $9 \times 10^{17} \text{ cm}^{-3}$; the CsPbI₃ absorber follows the dimethylammonium-iodide intermediate route with propyltriethoxysilane-assisted crystallisation that already delivered 22.60% in the laboratory [1], targeting 2.0–2.2 μm films with bulk lifetimes consistent with $N_t \sim 10^{12} \text{ cm}^{-3}$ (single-crystal-grade grains, verified by time-resolved photoluminescence). Interface passivation uses the fluorinated-dipole [2] or dual-interface co-passivation [3] chemistries to approach the modelled 10^{10} cm^{-2} state density, and CBTS is sputtered at 100 nm with $N_A \sim 10^{18} \text{ cm}^{-3}$ before Ni evaporation. In-line controls map directly onto the paper’s figures: QE edge at 752 nm (Fig. 5), Mott–Schottky depletion collapse near 0.5 V (Fig. 6), and the temperature coefficient from Table 7 ($-0.29\%/K$ in PCE over 300–400 K). Hitting all four simultaneously reproduces the 23.49% upper bound; missing any one lands on the corresponding contour of Fig. 3. The translation is summarised in Table 12.

11 Limitations and outlook

Three boundaries delimit this work. First, $N_t = 10^{12} \text{ cm}^{-3}$ at 2.2 μm demands near-single-crystal films; at the routinely achieved 10^{15} cm^{-3} our own map (Fig. 3) puts the optimum near 19% at sub-micron thickness, which is the honest near-term fab target. Second, phase stability of black CsPbI₃ against the yellow δ -phase, ion migration, and damp-heat degradation are outside SCAPS-1D steady-state scope and must be validated experimentally, following the passivation routes of [1–3]. Third, the single-junction ceiling ($\sim 30\%$ at 1.7 eV [31, 38, 39]) caps how far any honest optimisation can go; the credible path past 30% is a CsPbI₃/Si or all-perovskite tandem using our 1.65 eV recipe as the top cell ([12] demonstrates the bifacial/tandem methodology). The lead-free programme (CsSnI₃, double perovskites [11, 40]) can inherit the guardrailed factorial protocol directly.

12 Conclusion

Simultaneous factorial optimisation under enforced guardrails delivers a 23.49%-efficient CsPbI₃/CBTS cell that beats the certified experimental record with interfaces, parasitics accounting, and full reproducibility—and shows that the published 24%+ simulation values are interface-omission artefacts. We recommend guardrailed reporting (interface status, defect floors, unit-system declaration, Rs basis) as a publication standard for SCAPS studies.

Table 13: Simulation batches archived under `leadpaper/`.

Batch	Configurations	Receipt
Baseline validation	4 (one worker-crash retry)	<code>baseline_results.json</code>
Unit diagnostics	6 + 4 retries	<code>diag_units.json</code>
Round A factorial	96	<code>roundA.jsonl</code>
Round B refine	36	<code>roundB.jsonl</code>
Characterisation + audit + T	17 + 6 re-runs	<code>char.jsonl</code>

13 Data availability

All definition files (`csPbI3-CBTS-r7.def` with interfaces, `csPbI3-CBTS-noIF.def` without), sweep receipts (`roundA.jsonl`, `roundB.jsonl`, `char.jsonl`), raw curves (`champion_JV.json`), analysis (`analysis.json`) and figure scripts are archived with the project under `leadpaper/` (inventory in Table 13) (directories `defs/`, `outputs/`, `receipts/`, `logs/`). No intermediate data were held in temporary storage.

14 Author contributions

The authors designed the factorial study, built the SCAPS definition files, executed the simulation campaign, performed the audit and surrogate analysis, and wrote the manuscript.

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15 Appendix A: the scripting unit trap

SCAPS definition files store SI units (metres, m^{-3} , m^2/Vs) but script `set` commands expect practical units (μm , cm^{-3} , cm^2/Vs)—verified by dose-response: `set layer2.NA 1e15` reproduces the file baseline (stored 10^{21} m^{-3}) exactly, while SI-valued inputs (e.g. 10^{22}) inject a $10^6\times$ overdose and yield degenerate ghost cells (V_{oc} 1.47–1.53 V, FF 91–92%, up to 27%). Guardrail flags that catch them: $V_{\text{oc}} > E_g/q - 0.1 \text{ V}$, $\text{FF} > 90.5\%$, or $|\Delta|/\text{Vs}$ -thickness response inversion. All values in this paper are stated in the units the script layer expects; the archived logs record both.

16 Appendix B: run inventory

All simulation batches are inventoried in Table 13.

References

- [1] W. Dai, J. Li, Y. Gou, J. Zhang, Z. Pan, et al. Moisture-responsive crystallization strategy for efficient cspbi3 solar cells fabricated under high-humidity conditions. *Nature Communications*, 2026. 21.00% at 55% RH; 22.60% laboratory, 22.02% certified.

- [2] J. Qiu, X. Mei, M. Zhang, G. Wang, L. Pan, S. Zou, J. Huang, X. Zhang, and Y. T. Shi. Fluorination-assisted interfacial dipole for cspbi3 perovskite solar cells with over 22% efficiency. *Angewandte Chemie International Edition*, 137:e202506282, 2025.
- [3] N. Zhang, Z. Xu, Z. Yang, et al. Three birds with one stone: dual-interfaces and bulk co-passivation enable >21% efficiency of cspbi3 solar cells with voc of 1.27 v. *Science China Materials*, 69:405–413, 2026.
- [4] Liyan Hu, Chenzhe Xu, Haochun Sun, Hongwei Wang, Linkai Yu, Jinhao Yang, Feiyu Cheng, Zhuo Kang, and Yue Zhang. Pushing the performance of inverted cspbi3 perovskite photovoltaics via grain boundary grooves stitching. *Joule*, 2026. doi: 10.1016/j.joule.2026.102319. Record certified 21.43% inverted CsPbI3.
- [5] M. K. Hossain, G. F. I. Toki, I. Alam, R. Pandey, D. P. Samajdar, M. F. Rahman, M. R. Islam, M. H. K. Rubel, H. Bencherif, J. Madan, and M. K. A. Mohammed. Numerical simulation and optimization of a cspbi3-based perovskite solar cell to enhance the power conversion efficiency. *New Journal of Chemistry*, 2023. doi: 10.1039/d2nj06206b.
- [6] S. Mushtaq, S. Tahir, A. Ashfaq, R. S. Bonilla, M. Haneef, R. Saeed, W. Ahmad, and N. Amin. Performance optimization of lead-free masnbr3 based perovskite solar cells by scaps-1d device simulation. *Solar Energy*, 249:401–413, 2023. FTO/SnO2/MASnBr3/NiO/Au, 34.52% claimed; exceeds single-junction limit, used here for method only.
- [7] M. K. Hossain, M. H. K. Rubel, G. F. I. Toki, I. Alam, M. F. Rahman, and H. Bencherif. Effect of various electron and hole transport layers on the performance of cspbi3-based perovskite solar cells. *ACS Omega*, 7:43210–43230, 2022.
- [8] Nurul Farina Mohd Nazli, Fauziyah Salehuddin, Faiz Arith, Siti Aisah Mat Junos, Anis Suhaila Mohd Zain, and Khairil Ezwan Kaharudin. Numerical optimization of cspbi3 perovskite solar cells using scaps-1d modeling. In *2025 IEEE Regional Symposium on Micro and Nanoelectronics (RSM)*, pages 61–65, 2025. ITO/TiO2/CsPbI3/CBTS/Ni, 24.17% claimed.
- [9] S. O. Oyedele, I. Soumahoro, L. Bamba, R. Kre, and B. Aka. Simulation-based optimization of the performance of a cspbi3 (cesium lead triiodide) perovskite solar cell using scaps-1d. *Modeling and Numerical Simulation of Material Science*, 2023. ZnO/CsPbI3/PTAA, 24.24% claimed; no interface layers.
- [10] Mohammed Saleh Alshaikh. Perovskiteopt-ai: A machine learning-driven multi-parameter optimization framework for lead-free perovskite solar cell device architecture using scaps-1d simulation and gaussian process surrogate modeling. *Crystals*, 16:310, 2026. doi: 10.3390/cryst16050310. FTO/WS2/CsSnI3/CuSCN/Au, GP-BO to 27.83%.
- [11] Neda Nasiri, Seyed Mahdi Mastoor, and Amirhosein Ahmadkhan Kordbacheh. Multi-layer screening of double and conventional perovskite solar cells using scaps-1d and machine learning: Optimization of etl, htl, and absorber for high-efficiency architectures. *arXiv:2606.12083*, 2026. 125 architectures; Cs2AgInBr6 28.62% ML-suggested.

- [12] S. Rabhi, M. A. Benatallah, T. Hidouri, A. I. Shimul, B. C. Biswas, O. H. Alsalmi, H. A. Dmour, and M. W. Alam. Machine learning-enhanced scaps optimization of 28.38%-efficient bifacial triple-cation perovskite solar cells for building integration. *Surfaces and Interfaces*, 2026.
- [13] A. Swarnkar, A. R. Marshall, E. M. Sanehira, B. D. Chernomordik, D. T. Moore, J. A. Christians, T. Chakrabarti, and J. M. Luther. Quantum dot-induced phase stabilization of α -csppi3 perovskite for high-efficiency photovoltaics. *Science*, 354:92–95, 2016.
- [14] L. A. Frolova, D. V. Anokhin, A. A. Piryazev, S. Y. Luchkin, N. N. Dremova, K. J. Stevenson, and P. A. Troshin. Highly efficient all-inorganic planar heterojunction perovskite solar cells produced by thermal coevaporation of csi and pbi2. *Journal of Physical Chemistry Letters*, 8:67–72, 2017.
- [15] B. Zhao, S.-F. Jin, S. Huang, N. Liu, J.-Y. Ma, D.-J. Xue, Q. Han, J. Ding, Q.-Q. Ge, Y. Feng, and J.-S. Hu. Thermodynamically stable orthorhombic γ -csppi3 thin films for high-performance photovoltaics. *Journal of the American Chemical Society*, 140:11716–11725, 2018.
- [16] H. Bian, D. Bai, Z. Jin, K. Wang, L. Liang, H. Wang, J. Zhang, Q. Wang, and S. Liu. Graded bandgap csppi2+xbr1-x perovskite solar cells with a stabilized efficiency of 14.4%. *Joule*, 2:1500–1510, 2018.
- [17] J. Xi, C. Piao, J. Byeon, J. Yoon, Z. Wu, and M. Choi. Compositional engineering of α -csppi3 mixed-halide perovskites for efficient and stable solar cells. *Advanced Energy Materials*, 9:1901787, 2019.
- [18] P. Becker, J. A. Márquez, J. Just, A. Al-Ashouri, C. Hages, H. Hempel, M. Jošt, S. Albrecht, R. Frahm, and T. Unold. Low temperature synthesis of stable γ -csppi3 perovskite layers for solar cells obtained by high-throughput experimentation. *Advanced Energy Materials*, 9:1900555, 2019.
- [19] P. Luo, W. Xia, S. Zhou, L. Sun, J. Cheng, C. Xu, and Y. Lu. Solvent engineering for ambient-air-processed phase-stable csppi3 in perovskite solar cells. *Journal of Physical Chemistry Letters*, 7:3603–3608, 2016.
- [20] W. Ke, I. Spanopoulos, C. C. Stoumpos, and M. G. Kanatzidis. Myths and reality of hpbi3 in halide perovskite solar cells. *Nature Communications*, 9:4785, 2018.
- [21] S. Xiang, Z. Fu, W. Li, Y. Wei, J. Liu, H. Liu, L. Zhu, R. Zhang, and H. Chen. Highly air-stable carbon-based α -csppi3 perovskite solar cells with a steady-state efficiency of 9.5%. *ACS Energy Letters*, 3:1824–1831, 2018.
- [22] Y. Wang, X. Liu, T. Zhang, X. Wang, M. Kan, J. Shi, and Y. Zhao. Aziridinium lead iodide for efficient and stable csppi3 perovskite solar cells. *Angewandte Chemie International Edition*, 58:16691–16696, 2019. 18.40% experimental CsPbI3.
- [23] K. D. Jayan and V. Sebastian. Comprehensive device modelling and performance analysis of masni3 based perovskite solar cells with diverse etm, htm and back metal contacts. *Solar Energy*, 221:99–108, 2021. FTO/ZnO/CsPbI3/CuSbS2/Se, 15.60% theory.

- [24] S. E. Boussaada, Y. Mouchaal, H. Riane, and A. Khelil. Deep insights into the coupled optoelectronic analysis of etl thin films and photovoltaic analysis of cspbi3-based perovskite solar cell using scaps-1d simulations. *Preprint (exa.ai record 1ptgsxrjmc)*, 2025. FTO/SnCoOx/CsPbI3/Cu2O/Au, 21.34%.
- [25] M. K. Ristiansyah, B. Prasetya, M. G. I. Khan, M. Y. Siregar, A. A. S. Panjaitan, E. Widianto, et al. Design and optimization of cspbi3 perovskite solar cells using scaps-1d. *Greenusmater*, 2026. FTO/ZnO/CsPbI3/Spiro-OMeTAD, 16.3 to 23.1%.
- [26] C. Son, H. Son, and B.-S. Jeong. Enhanced conversion efficiency in mapbi3 perovskite solar cells through parameters optimization via scaps-1d simulation. *Applied Sciences*, 14:2390, 2024. ITO/TiO2/MAPbI3/Cu2O/Au, 19.30%.
- [27] R. Ritu, G. Gagandeep, R. Kumar, and F. Chand. Performance optimization of fapbi3 based perovskite solar cell. *AIP Conference Proceedings*, 2995:020142, 2024. FTO/TiO2/FAPbI3/Spiro-OMeTAD/Ag, 20.34% with gradient doping.
- [28] H. Malla and O. Onay. Investigation of the effect of different factors on the performance of several perovskite solar cells: a simulation study by scaps. *Preprint (exa.ai record lbp26v9nb9y)*, 2022. MAPbI3 24.8%, FAPbI3, CsPbI3 and lead-free set with TiO2/Cu2O.
- [29] W. M. Mohamad. Strategic sto/cusen material integration for 31.49% efficient fapbi3 perovskite device performance through computational simulation. *Preprint (exa.ai record jmxzvkkzwt3)*, 2025. Glass/ITO/STO/FAPbI3/CuScN/Au, 31.49% with back-surface layer.
- [30] M. A. Rahman and M. J. Alam. Machine learning-guided scaps-1d design of lead-free ch3nh3snI3 perovskite solar cells with 40.17% simulated pce. *Journal of Materials Science: Materials in Engineering*, 2026. Idealized upper limit exceeding the Shockley-Queisser bound.
- [31] W. Shockley and H. J. Queisser. Detailed balance limit of efficiency of p-n junction solar cells. *Journal of Applied Physics*, 32:510–519, 1961.
- [32] Y. H. Khattak. *Modeling of High Power Conversion Efficiency Thin Film Solar Cells*. PhD thesis, Universitat Politecnica de Valencia, 2019. SCAPS device physics reference.
- [33] M. Burgelman, K. Decock, A. Niemegeers, J. Verschraegen, and S. Degrave. Scaps-1d solar cell capacitance simulator (version 3.3.10), 2021. Device simulator and scripting reference.
- [34] Y. Raoui, H. Ez-Zahraouy, N. Tahiri, O. El Bounagui, S. Ahmad, and S. Kazim. Performance analysis of mapbi3 based perovskite solar cells employing diverse charge selective contacts: Simulation study. *Solar Energy*, 193:948–955, 2019.
- [35] M. H. Azar, S. Ayneband, H. Abdollahi, H. Alimohammadi, N. Rajabi, S. Angizi, et al. Scaps empowered machine learning modelling of perovskite solar cells: Predictive design of active layer and hole transport materials. *Preprint (exa.ai record 5p4qq4j6qwd)*, 2023. FTO/SnO2/FAPbI3/CsSnI3/Au, 23.94%.
- [36] B. Li, Y. Zhang, L. Fu, T. Yu, S. Zhou, L. Zhang, and L. Yin. Surface passivation engineering toward high efficiency perovskite solar cells. *Nature Communications*, 9:1076, 2018.

- [37] Ivan E. Novoselov, Alexander M. Gvozdev, Andrey A. Smirnov, and Ivan S. Zhidkov. Dataset of scaps-1d simulated halide perovskite solar cells with shap and machine learning-based pce optimization. *Data in Brief*, 60:111653, 2025. doi: 10.1016/j.dib.2025.111653. 7182 configurations; Zenodo 10.5281/zenodo.15211402.
- [38] M. A. Green, E. D. Dunlop, J. Hohl-Ebinger, M. Yoshita, N. Kopidakis, and A. W. Y. Ho-Baillie. Solar cell efficiency tables (version 66). *Progress in Photovoltaics*, 33:795–810, 2025.
- [39] National Renewable Energy Laboratory. Best research-cell efficiency chart, 2026. Accessed 2026; perovskite/Si tandems 33–34%, single-junction perovskite $\sim$ 26–27%.
- [40] Parisa Karimi Moone and Nafiseh Sharifi. Comparison of pb-based and sn-based perovskite solar cells using scaps simulation: optimal efficiency of eco-friendly csni3 devices. *Environmental Science and Pollution Research*, 31:51447–51460, 2024. doi: 10.1007/s11356-024-34622-x. CsSnI3 17.36% optimized.